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Classical vs. Learned Approaches to Relative Pose Estimation

A Comparative Study of Classical Geometric Estimation and SRPose on TUM RGB-D
Data

Internship Report

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1 Introduction

Relative pose estimation refers to the problem of recovering the rigid transformation, a rotation and a translation, between the viewpoints of two cameras observing a common scene, given two images captured from those viewpoints. It is a foundational problem in computer vision, underlying applications such as visual odometry, Structure-from-Motion, simultaneous localization and mapping (SLAM), and augmented reality [6]. Two broad paradigms exist for solving Relative Pose Estimation: Classical approaches and deep-learning-based approaches.

Classical geometry-based approaches are grounded in the projective geometry of two-view systems and in statistical models of noise measurement, and they proceed through a well-defined sequence of stages. First, distinctive keypoints are detected in each image, and correspondences are established by matching their descriptors across the image pair. From these correspondences, outliers are filtered out, and a fundamental matrix F is estimated in a way that best explains the remaining correspondences. The fundamental matrix is a 3×3 rank-2 matrix that encodes the epipolar constraint between the two views: for any pair of corresponding points x and x' in the two images, $x'^T F x = 0$ [6, 10]. Geometrically, F maps a point in one image to its corresponding epipolar line in the other, along which the true match must lie [7]. Given known camera intrinsics K_1 and K_2 , F can be converted into the essential matrix $E = K_2^T F K_1$ [6], from which the relative rotation and translation between the two views can be recovered.

Deep-learning-based approaches, by contrast, replace some or all of these geometric stages with an end-to-end trainable model. A typical architecture takes the two images directly as input, extracts keypoints and their descriptors using a learned feature extractor, and then processes these features through cross-attention layers that model relationships between keypoints across and within the two images. Rather than estimating F or E explicitly, such models directly regress the relative pose from these learned feature relationships [17].

The relevance of classical approaches lies precisely in its explicit geometric and statistical grounding: by modeling the noise properties of correspondence data directly, methods such as Kanatani’s maximum-likelihood estimator [8] aim to produce estimates that are optimal given an assumed noise model: a property that learned methods, trained to minimize an aggregate loss over a large dataset, do not explicitly guarantee for any individual input. This raises a natural question: given that learned methods such as SRPose now constitute the state of the art in relative pose estimation, can a carefully implemented and statistically principled classical pipeline match or exceed their precision, and if so, under what circumstances does each approach have the advantage? This work investigates that question empirically, comparing a full classical pipeline against SRPose on real-world RGB data.

2 Literature Review

2.1 Correspondence Extraction: SIFT

As a first step of the classical method for relative pose estimation, the aim is to extract matching points between two views of an image pair, and this is exactly what SIFT does.

SIFT (Scale-Invariant Feature Transform) is a method for detecting and describing local features in images, originally introduced by Lowe [11]. It operates in four main stages: Detecting local extrema of a DoG function (Difference of 2 Gaussian-convolved images) as candidate keypoints, localizing keypoints and refining them by fitting a detailed model, assigning orientations to each keypoint from local image gradient directions in its neighborhood, and computing local descriptor for each keypoint from its local image gradients [11]. Correspondences between two images are

then established by comparing the descriptor vectors of keypoints extracted independently from each image, via nearest-neighbor search in descriptor space, with a ratio test (Section 3.2.1) used to reject ambiguous matches.

SIFT was selected here for several reasons. First, it is scale and rotation invariant, as well as robust to viewpoint change, illumination variation, and noise [11]. This makes it well matched to the real-world imaging conditions present in the RGB dataset used for evaluation. Moreover, SIFT remains a standard, extensively validated baseline in the classical feature-matching literature, with well-understood behavior and failure modes, which preserves a cleaner methodological separation between the classical and learned approaches under comparison.

2.2 Fundamental Matrix Estimation

2.2.1 Eight-Point Algorithm

Given a set of normalized correspondences, a candidate fundamental matrix is estimated via the eight-point algorithm [7, 10]. Each correspondence contributes one row to a linear system $Af = 0$, where f is the vectorized fundamental matrix [10]. A first estimate is obtained as the null-space vector of A via singular value decomposition (SVD), and the rank-2 constraint required of a valid fundamental matrix is then enforced by setting the smallest singular value of this initial estimate to zero and reconstructing the matrix.

2.2.2 RANSAC and AC-RANSAC

RANSAC (Random Sample Consensus), introduced by Fischler and Bolles [4], is a robust estimation framework designed to fit a model to data contaminated by outliers. Applied to fundamental matrix estimation, RANSAC operates by repeatedly drawing a minimal random subset of correspondences (eight, in the case of the linear eight-point algorithm [7]), fitting a candidate fundamental matrix to that subset, and evaluating how many of the remaining correspondences are consistent with the resulting model within some error tolerance threshold. Correspondences whose residual error falls below this tolerance are counted as **inliers**. This process is repeated for a number of iterations, adaptively reduced as larger inlier sets are found [4], and the candidate model (Fundamental Matrix in our case) with the largest inlier support is retained, typically followed by a final re-estimation using all identified inliers.

AC-RANSAC, also known as ORSA (Optimized Random Sampling Algorithm), was introduced by Moisan and Stival [13] as a variant of RANSAC that eliminates the need for a manually specified error threshold. It is grounded in the a contrario methodology [3], in which a candidate model is evaluated by the number of false alarms (NFA) it would be expected to generate under a null hypothesis of random and uniformly distributed correspondences. A model is accepted only if this NFA is sufficiently small and the precision is determined automatically as part of this optimization [13]. This approach was later refined and generalized by Moisan, Moulon, and Monasse [12], who formalized the underlying automatic RANSAC framework and its extension to fundamental matrix estimation with a contrario elimination of outliers.

2.2.3 Kanatani’s Maximum Likelihood F Estimation

The Fundamental Matrix can also be estimated using Kanatani’s Maximum Likelihood model [8].

Setting. We begin with the epipolar constraint, expressed using Kanatani’s homogeneous coordinate convention with a conditioning constant f_0 used to scale the coordinates depending on the

image size:

$$\begin{pmatrix} x \\ y \\ f_0 \end{pmatrix}^\top \begin{pmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{pmatrix} \begin{pmatrix} x' \\ y' \\ f_0 \end{pmatrix} = 0, \quad \mathbf{x}^\top F \mathbf{x}' = 0, \quad \mathbf{x} = (x, y, f_0)^\top, \quad \mathbf{x}' = (x', y', f_0)^\top. \quad (1)$$

We vectorize both the fundamental matrix and the point correspondence coordinates as:

$$u = (F_{11}, F_{12}, F_{13}, F_{21}, F_{22}, F_{23}, F_{31}, F_{32}, F_{33})^\top, \quad (2)$$

$$\xi = (xx', xy', xf_0, yy', yy', yf_0, f_0x', f_0y', f_0^2)^\top, \quad (3)$$

which allows the epipolar constraint (Eq. 1) to be rewritten simply as $(u, \xi) = 0$.

The goal is to find the vector u that satisfies this equation for *all* point correspondences.

Noise model. The difficulty is that point correspondences are noisy. Writing ξ_α in terms of its noise-free value and an additive perturbation, we get: $\xi_\alpha = \bar{\xi}_\alpha + \Delta\xi_\alpha$. Kanatani [8] introduces the covariance matrix $V[\xi_\alpha]$, which measures the variance of each component of ξ_α together with the covariance between every pair of its components.

Hypothesis: each image coordinate of a matching point pair is perturbed by independent random noise; x , x' , y , and y' are independent random variables of variance σ^2 , perturbed by independent noise of mean 0 and standard deviation σ . From that, he decomposes $V[\xi_\alpha]$ as: $V[\xi_\alpha] = \sigma^2 V_0[\xi_\alpha]$, with $V_0[\xi_\alpha]$ being a 9×9 matrix depending only on the coordinates $x_\alpha, y_\alpha, x'_\alpha, y'_\alpha$ and the conditioning constant f_0 .

Maximum likelihood estimation. Assuming no outliers, each ξ_α is treated as an independent Gaussian random variable with mean $\bar{\xi}_\alpha$ and covariance matrix $V[\xi_\alpha]$. Maximum likelihood estimation of u then amounts to minimizing the sum of squared Mahalanobis distances of the data points ξ_α to the hyperplane in $\mathbb{R}^9$ with normal u , which means minimizing the discrepancy between the observed and true ξ_α :

$$J_{\text{ML}} = \frac{1}{2} \sum_{\alpha=1}^N (\xi_\alpha - \bar{\xi}_\alpha, V_0[\xi_\alpha]^{-1} (\xi_\alpha - \bar{\xi}_\alpha)), \quad (4)$$

subject to the constraint $(u, \bar{\xi}_\alpha) = 0$. In Eq. 4, each squared deviation $\xi_\alpha - \bar{\xi}_\alpha$ is weighted by the inverse of $V_0[\xi_\alpha]$: a point that deviates from its true value in a direction of high expected noise contributes little to the cost, whereas a deviation in a direction of low expected noise, where the correspondence is inherently more reliable, is penalized more heavily.

After introducing Lagrange multiplier, we obtain the following equation which we aim to minimize (The details of this derivation can be found in the Appendix Section A.2):

$$J_{\text{ML}} = \frac{1}{2} \sum_{\alpha=1}^N \frac{(u, \xi_\alpha)^2}{(u, V_0 u)}, \quad (5)$$

The goal becomes finding u that minimizes Eq. 5. Differentiating with respect to u gives:

$$\nabla_u J_{\text{ML}} = \sum_{\alpha=1}^N \frac{(u, \xi_\alpha) \xi_\alpha}{(u, V_0[\xi_\alpha] u)} - \sum_{\alpha=1}^N \frac{(u, \xi_\alpha)^2 V_0[\xi_\alpha] u}{(u, V_0[\xi_\alpha] u)^2} = (M - L) u = 0. \quad (6)$$

where $M = \sum_{\alpha=1}^N \frac{\xi_\alpha \xi_\alpha^\top}{(u, V_0[\xi_\alpha] u)}$, and $L = \sum_{\alpha=1}^N \frac{(u, \xi_\alpha)^2 V_0[\xi_\alpha]}{(u, V_0[\xi_\alpha] u)^2}$.

FNS. The Fundamental Numerical Scheme (FNS), introduced by Chojnacki et al. [2], solves Eq. 6 iteratively: at first vector u is initialized, and at each iteration, the matrices M and L are recomputed from the current estimate of u . u is updated as the eigenvector of $M - L$ associated with the eigenvalue closest to zero, with the process repeated until convergence. Because $(M - L)u = 0$ holds only at a stationary point of J_{ML} , this eigenvector-based update scheme allows FNS to converge to a solution that directly satisfies the maximum-likelihood optimality condition, rather than relying on a fixed linear solve.

Limitations of the eight-point algorithm and motivation for Kanatani’s method. The classical eight-point algorithm [7, 10] treats every correspondence as contributing equally to the least-squares fit $Af = 0$. This implicitly assumes that all correspondences carry the same, isotropic level of noise, regardless of their position in the image, which does not hold in general. Kanatani’s maximum-likelihood formulation directly incorporates the heteroscedastic noise structure of correspondences through $V_0[\xi_\alpha]$, yielding an estimate that is statistically more efficient than the unweighted eight-point solution.

2.3 Deep Learning for Relative Pose Estimation - SRPose

In recent years, deep learning has emerged as a competitive alternative to classical, correspondence-based geometric estimation for the relative pose estimation problem. A range of learned approaches have been proposed, spanning direct pose regression networks that predict (R, t) end-to-end from a pair of images [9], learned feature-matching pipelines such as SuperGlue [14] and LoFTR [16], and more recent transformer-based architectures that unify feature extraction, matching, and pose regression within a single end-to-end trainable model [17]. Among these, **SRPose** [17] was selected as the learned baseline for this work, as it represents a recent, state-of-the-art architecture that addressed two limitations of previous learned pose regressors: sensitivity to varying camera intrinsics across image pairs, and lack of generalizability across image resolutions and scene types.

2.3.1 SRPose Architecture

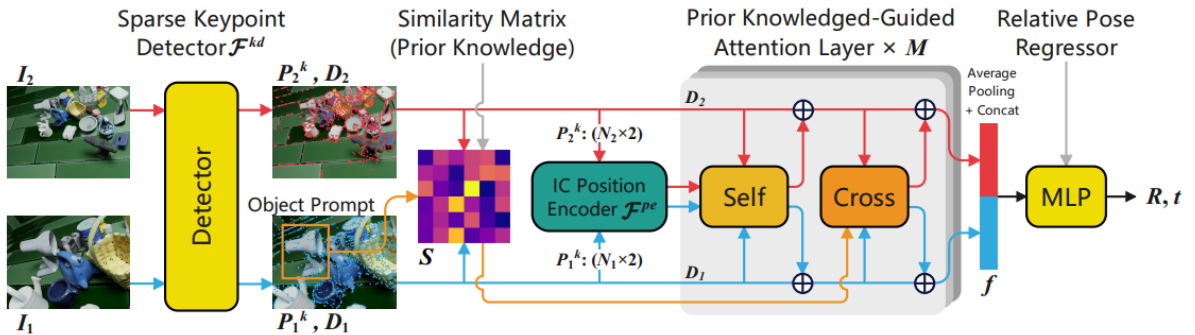


Figure 1: SRPose Architecture.[17]

In SRPose, the image pair I_1 and I_2 is passed to a pre-trained sparse keypoint detector, which extracts a small number of informative keypoints instead of dense image features. Each keypoint has an associated descriptor, confidence score, and location. The keypoints are then passed to an intrinsic-aware positional encoder: Previous regressors extract implicit image features with encoders trained on fixed-size images, assuming that the camera intrinsics of all inputs are the same [17].

In SRPose, by contrast, the coordinates of the keypoints are first normalized using the camera intrinsics (K_1 and K_2), via $K^{-1} \cdot [u, v, 1]^\top$, before applying the standard positional encoding. This means that points lying on the same physical ray will receive the same positional encoding regardless of the intrinsics of the camera that observed them, enabling generalization across different cameras [17].

For each image, the keypoint descriptor embeddings, with the intrinsic-aware position embedding added, are fed into the attention layers. Each attention layer is composed of a self-attention module and a cross-attention module.

Prior knowledge-guided cross-attention module: The cross-attention process is guided by a prior, a similarity matrix S computed between the descriptors of the keypoints in I_1 and I_2 : $S(i, j) = d_i \cdot d_j$. This bias term multiplies the cross-attention score, encouraging the transformer to attend to geometrically plausible correspondences without ever committing to hard matches, which makes it robust to repetitive textures, partial occlusions, and viewpoint changes.

Finally, the embeddings of the attention layers are passed to a pose regressor (an MLP), and both rotation and translation are regressed jointly in a single forward pass.

Further architectural detail, including the cross-attention formulation and full loss function, is provided in Appendix B

3 Method

3.1 Pipeline Overview

We first implemented a pipeline for relative pose estimation, following the steps in Figure 2. From an image pair, we implemented SIFT for keypoint detection and matching. We implemented RANSAC and AC-RANSAC for extracting inliers from the matches and for estimating the fundamental matrix using the eight-point algorithm. Upon that, we implemented Kanatani’s Fundamental Matrix estimation method FNS to refine RANSAC/AC-RANSAC’s F which was used as an initial guess. From FNS’ F matrix and using the intrinsics of the images, we built the Essential Matrix E , which we then decomposed using SVD into 4 rotation and translation (R, t) candidates. The final solution was selected by triangulating the inliers and choosing the candidate for which the largest number of 3D points lie in front of both cameras (cheirality test).

3.2 Correspondence Extraction

Correspondences between each image pair are established using the Scale-Invariant Feature Transform (SIFT) [11], as described in Section 2.1.

3.2.1 Nearest-Neighbor Matching and Ratio Test in SIFT

For each keypoint in the first image, its two nearest neighbors in descriptor space are identified among the keypoints of the second image, using squared Euclidean distance between descriptor vectors. A candidate match is accepted only if it passes Lowe’s ratio test [11]: denoting the squared distance to the nearest and second-nearest descriptor as d_1 and d_2 respectively, a match is retained only if

$$d_1 < \tau^2 \cdot d_2, \quad (7)$$

where τ is the ratio threshold.

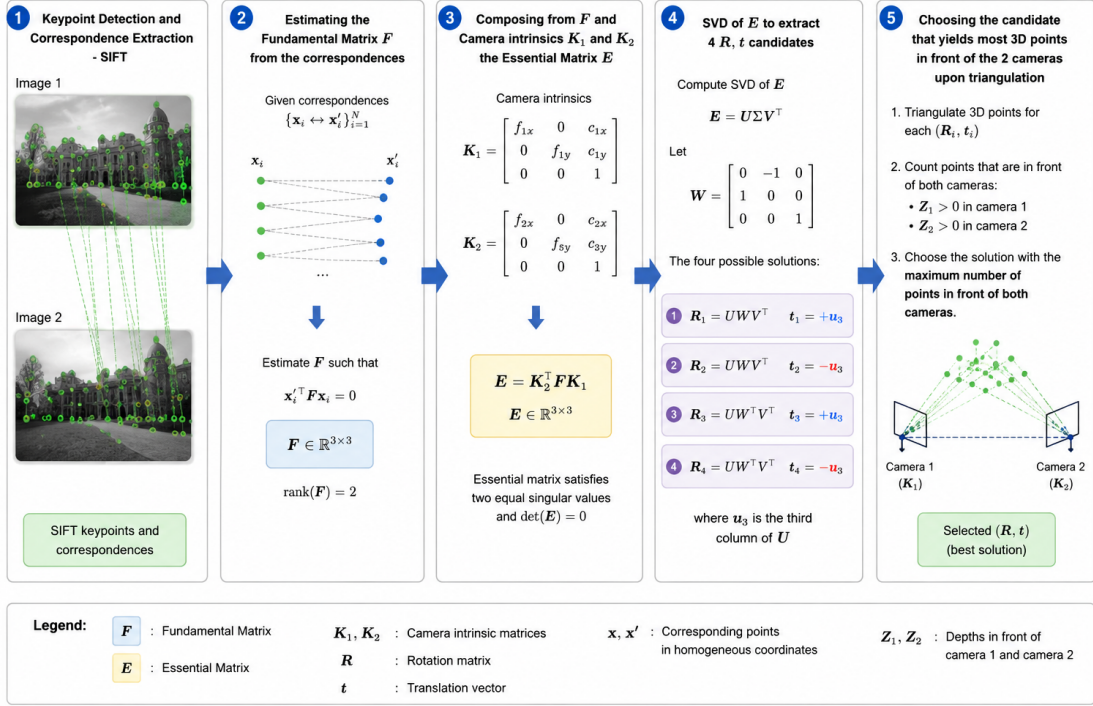


Figure 2: Classical Pipeline for relative pose estimation from two images.

A ratio threshold of $\tau = 0.6$ was used throughout this work, chosen within the range typically recommended in the literature [11]. This test rejects ambiguous matches, retaining only matches for which the best candidate is a much better fit than any alternative.

3.2.2 Lens Distortion Correction

The pipeline then continues and carries out an "undistortion" process. Since we will be testing on real data taken from physical cameras with real lenses, the projection of 3D points onto the image plane is affected by lens distortion (radial and tangential distortion). Because our classical methods rely on epipolar geometry, which assumes that the image points are observed according to the ideal camera projection model where no distortion exists, undistortion is necessary.

Given a distorted pixel coordinate, the corresponding undistorted normalized coordinate is recovered via fixed-point iteration, and the result is projected back into pixel space using the same intrinsic matrices. This correction is applied directly to inliers of both images of every pair, using the distortion parameters of the camera used to capture the sequence.

3.3 Fundamental Matrix Estimation

3.3.1 Normalized Eight-Point Algorithm, RANSAC, and AC-RANSAC

First, the pixel coordinates of each set of correspondences are normalized following the isotropic scaling procedure of Hartley [7], which is essential to the numerical conditioning of the subsequent linear estimation step. For each image independently, the correspondence coordinates are translated so that their centroid lies at the origin, and uniformly scaled so that their average distance from the origin equals $2\sqrt{2}$. This yields two 3×3 normalization matrices, N_1 and N_2 , applied

respectively to the correspondences of the first and second image. After that, the Eight-Point Algorithm described in Section 2.2.1 is applied, giving us an estimate of the Fundamental Matrix F . Finally, the estimate is denormalized via $F = N_2^\top F_n N_1$, where F_n denotes the fundamental matrix in normalized coordinates, giving a fundamental matrix in pixel space and satisfying the epipolar constraint: $x_2^\top F x_1 = 0$.

We then implemented both RANSAC and AC-RANSAC: In the standard RANSAC implementation [4], the inliers are identified as correspondences whose Sampson distance [6] to the candidate model falls below a fixed threshold, $\delta_{\max} = 1.5$ pixels. In AC-RANSAC (ORSA) [13], the inlier threshold is determined automatically for each image pair via an a-contrario criterion.(Section 2.2.2).

3.3.2 Refinement via Kanatani’s FNS

To apply Kanatani’s Fundamental matrix estimation, we implemented all the estimators mentioned in Section A.4: FNS, HEIV, Renormalization, and Gauss-Newton. However, only FNS was selected for evaluation on real images since it gave the best convergence behavior on synthetic data.

FNS requires an initial estimate u_{init} of the vectorized unit-norm fundamental matrix; this is obtained from the RANSAC/AC-RANSAC output by normalizing it to unit Frobenius norm, converting it from the pixel-space convention into Kanatani’s f_0 -conditioned convention, and transposing it since FNS solves $x_1^\top F x_2 = 0$ while the eight-point algorithm solves $x_2^\top F x_1 = 0$. f_0 is fixed at 600, an order-of-magnitude matching the image dimensions used throughout this work.

FNS is then run for a maximum of 100 iterations, or until successive estimates of u agree to within a fixed tolerance, and the resulting fundamental matrix is de-normalized and returned in the standard pixel-space convention used throughout the remainder of the pipeline.

3.4 Essential Matrix Recovery and Pose Decomposition

Given the camera intrinsics K_1 and K_2 and the refined fundamental matrix F , the essential matrix is computed as $E = K_2^\top F K_1$. E is then normalized to unit Frobenius norm, decomposed via SVD, and enforced to rank-2 structure by averaging its two largest singular values and setting the smallest to zero.

A second SVD of the corrected E yields U, V , from which four candidate poses $(R_1, t_1), (R_1, -t_1), (R_2, t_1)$, and $(R_2, -t_1)$ are formed via $R = UWV^\top$ (using the standard rotation matrix W) and $t = U_{:,3}$ [6].

The correct candidate is selected via the cheirality test: each candidate is used to triangulate all inlier correspondences, and the candidate for which the largest number of points have positive depth in both cameras is selected. When the top two candidates’ vote counts differ by less than a fixed margin, the tie is broken using total reprojection error across all correspondences, favoring the candidate with the lower average error.

3.5 Validation Methodology: Synthetic Testing

As a validation step of the pipeline created, and to isolate the environment from any additional and noisy factors that may be present in real image data, we ran the pipeline on synthetic data.

The synthetic camera intrinsics were set to $f_x = f_y = 500$, $c_x = 320$, $c_y = 240$, with $K_1 = K_2$. The ground-truth rotation was constructed from Euler angles of 0.6, 0.3, and 0.2 radians (yaw, pitch, and roll respectively, corresponding to a combined rotation magnitude of approximately 34°), and the ground-truth translation was set to $t = (0.5, 0.1, -0.2)$. A set of random 3D points is generated and projected noiselessly into both views, and the resulting exact correspondences

are passed through the full pipeline 2, from fundamental matrix estimation to final pose recovery which can be compared to the ground-truth pose generated.

Since both the input correspondences and the ground-truth pose are known exactly and no noise can be present, this procedure was used throughout development to verify each stage of the pipeline prior to evaluation on a real dataset.

3.6 TUM RGB-D Dataset, Camera Model and Ground Truth: Real-Image Testing

To test on real images, the TUM RGB-D dataset was chosen [15]. It is a widely used benchmark for evaluating visual odometry, SLAM, and relative pose estimation methods, providing synchronized RGB and depth image sequences captured with a Microsoft Kinect sensor, together with high-frequency, high-precision ground-truth camera trajectories recorded by an external motion-capture system. Among the different calibration groups, we chose the dataset *Freiburg1* (fr1) (recorded at Freiburg’s computer vision laboratory), in particular the *fr1/room* sequence, which reports 1000 RGB frames captured [15]. It is obtained by a handheld trajectory recorded through a full office room, and it was specifically chosen for its greater scene and depth diversity. This sequence also comes with its own published camera intrinsics (f_x, f_y, c_x, c_y) and radial-tangential lens distortion coefficients (k_1, k_2, k_3, p_1, p_2)[15].

Since RGB frames and ground-truth pose annotations are recorded asynchronously and are not guaranteed to share identical timestamps, each RGB frame must be matched to its closest corresponding ground-truth pose entry. This association is performed using the `associate.py` script provided by TUM alongside the dataset [15], which pairs each RGB frame with the ground-truth pose entry whose timestamp is closest in absolute value, subject to a maximum allowed time difference. The resulting association is used throughout this work both to construct ground-truth relative poses between frame pairs and to identify the corresponding RGB image files for a given pair index.

3.7 Evaluation Metrics

Rotation Error: Given a predicted rotation R_{pred} and ground-truth rotation R_{gt} , the rotation error is computed as the geodesic angle between them: $\theta_R = \arccos\left(\frac{\text{tr}(R_{\text{gt}}^\top R_{\text{pred}}) - 1}{2}\right)$, reported in degrees.

Translation error: Since translation is only recoverable up to an unknown scale from monocular relative pose estimation, translation error is measured as the angle between the predicted and ground-truth translation directions: $\theta_t = \arccos\left(\frac{t_{\text{gt}} \cdot t_{\text{pred}}}{|t_{\text{gt}}| \cdot |t_{\text{pred}}|}\right)$, also reported in degrees.

Both the mean and median of rotation and translation error are reported across all evaluated pairs for a given configuration.

Essential matrix comparison: To assess the accuracy of the estimated essential matrix independently of the final decomposed pose, the normalized Frobenius distance between the estimated E and the ground-truth $E_{\text{gt}} = [t_{\text{gt}}]_{\times} R_{\text{gt}}$ is computed, with both matrices normalized to unit Frobenius norm prior to comparison to account for the inherent scale ambiguity of E . The smaller of the two possible sign alignments is then reported: $d(E, E_{\text{gt}}) = \min\left(\|\hat{E} - \hat{E}_{\text{gt}}\|_F, \|\hat{E} + \hat{E}_{\text{gt}}\|_F\right)$, which accounts for the fact that E and $-E$ represent the same epipolar geometry.

Epipolar distance: The self-consistency of an estimated fundamental matrix F with its correspondences is measured via the Sampson distance [6] between each correspondence and the epipolar line predicted by F , averaged over all inlier correspondences.

Match coverage: To assess the spatial distribution of correspondences across the image, independent of their raw count, a grid-based coverage metric is computed for each image pair. The image is divided into an $N \times N$ grid (with $N=8$), and the fraction of grid cells containing at least one matched keypoint is reported as the coverage of that pair. This metric complements raw inlier count as a diagnostic.

3.8 SRPose Implementation

SRPose was evaluated using the official implementation and pretrained model weights released by the authors on GitHub [18], following the usage instructions and implementation details provided in the repository. The required Python dependencies were installed from the repository’s `requirements.txt` file, and inference was performed using the ScanNet-pretrained checkpoint.

It is important to note that SRPose was trained on multiple datasets: ScanNet, MegaDepth, HO3D, Linemod, etc. After checking the documentation of some of them, we noted that MegaDepth dataset was fully undistorted before being fed to SRPose for training [1].

3.9 Experimental Setup

3.9.1 Frame Pairing

Evaluation was performed on the *fr1/room* sequence of the TUM RGB-D dataset. Image pairs were formed by selecting, for a given *frame-skip* value s , the frame at association index i and the frame at index $i+s$.

frame-skip values of $s \in \{2, 5, 10, 12\}$ were evaluated, each spanning a different range of relative camera motion. Rather than exhaustively evaluating every possible starting index, pairs were sampled with a fixed stride of 5 across the association sequence to reduce the substantial overlap between consecutive candidate pairs while retaining a representative sample of several hundred pairs per frame-skip value.

3.9.2 Output Format

For each configuration (frame-skip value), results are written to two output files: a per-pair CSV file, recording the pair index, image filenames, ground-truth squared translation magnitude, ground-truth rotation magnitude, frame-skip value, inlier count, match coverage, estimation method (FNS in our case), and rotation and translation error; and a summary text file, recording the method, frame-skip value, and the mean and median rotation and translation error across the configuration. The pre-trained SRPose model was directly run on the same TUM RGB-D image pairs with the same frame-skip values, and results were recorded in the same per-pair CSV and summary text file format, omitting the inlier count and match coverage fields, which are not defined for SRPose’s correspondence-free architecture.

3.10 Code and Data Availability

The full source code for the classical pipeline implemented from scratch using C++ is publicly available at [GitHub](#). The implementation of SRPose model (inference only), the *fr1/room* sequence, and the generated result files of both methods as well as the code generating the plots and figures used in this work are available in this shared [OneDrive folder](#).

4 Results

4.1 Synthetic Validation of the Classical Pipeline

Before evaluating the classical pipeline on real TUM RGB-D data, we first verified its correctness end-to-end using the synthetic testing procedure described in Section 3.5.

We can see in Table 1 that the Fundamental matrix matched the inliers precisely with a near zero mean epipolar distance, and that the estimated essential matrix matched the ground-truth essential matrix to within numerical precision. This confirms that fundamental matrix estimation, essential matrix recovery, and cheirality-based pose disambiguation are each implemented correctly and recover the ground-truth pose exactly under noiseless conditions.

Table 1: Synthetic validation results for the classical pipeline, using the original RANSAC backend.

Quantity	Value
Inliers retained	170 / 170
Estimated precision (σ) / threshold	fixed $\delta_{\max} = 1.5$ px
E singular values (before correction)	0.7071, 0.7071, 7.4×10^{-20}
$\ E_{\text{estimated}} - E_{\text{true}}\ $ (before correction)	8.61×10^{-11}
$\ E_{\text{estimated}} - E_{\text{true}}\ $ (after correction)	8.59×10^{-11}
Mean epipolar (Sampson) distance on inliers	1.78×10^{-10} px

4.2 Improving the Classical Pipeline: RANSAC vs. AC-RANSAC (ORSA)

Table 2 compares the classical pipeline’s rotation and translation error using the original RANSAC backend against the ORSA (AC-RANSAC) backend, across four frame-skip values on real TUM RGB-D sequences.

Median rotation error favors ORSA at three of four skip values (5, 10, and 12), with the two backends within 0.02° of each other at skip = 5. Mean rotation error shows a less consistent pattern: ORSA outperforms RANSAC at skip = 2 and 5, while RANSAC achieves lower mean rotation error at skip = 10 and 12. Translation error favors ORSA at nearly every skip-metric combination, with the sole exception of median translation error at skip = 12, where the two backends are within 0.3° of each other.

For that reason, the pipeline using ORSA as backend was used to compare performance with SRPose.

Table 2: Mean and median rotation and translation errors for the classical pipeline using the RANSAC backend versus the AC-RANSAC backend, across four frame-skip values.

Skip	Backend	Mean Rot. Err. ($^\circ$)	Median Rot. Err. ($^\circ$)	Mean Trans. Err. ($^\circ$)	Median Trans. Err. ($^\circ$)
2	RANSAC	0.782	0.652	54.762	52.098
	ORSA	0.730	0.586	48.892	44.199
5	RANSAC	6.389	1.252	49.602	42.324
	ORSA	3.884	1.270	48.475	46.834
10	RANSAC	12.429	2.443	50.019	46.979
	ORSA	16.718	2.072	47.212	42.250
12	RANSAC	15.743	2.626	47.984	37.022
	ORSA	14.669	2.492	45.894	37.337

4.3 Aggregate Comparison: Classical Method vs. SRPose

Table 3 compares SRPose and the classical pipeline’s mean and median rotation and translation errors across four skip values on TUM data. SRPose achieves lower mean and median error than the classical pipeline for both rotation and translation at every tested skip value. The gap between mean and median rotation error is substantially larger for the classical pipeline than for SRPose, and this gap widens with increasing skip: at skip = 2, classical’s mean-median difference for the rotation error is 0.14° , growing to 16.24° at skip = 12, while SRPose’s corresponding gap remains under 0.2° across all four skip values.

This means that the classical pipeline’s error distribution becomes increasingly right-skewed at larger skip values, whereas SRPose’s error distribution remains comparatively symmetric and stable.

To check the reason of the distribution skew and to inspect possible outliers, we plotted the histograms of rotation errors of the 200 image pairs for both the classical method and SRPose per each skip value.

Table 3: Mean and median rotation and translation errors for the classical (ORSA-based) pipeline versus SRPose, across four frame-skip values.

Skip	Method	Mean Rot. Err. ($^\circ$)	Median Rot. Err. ($^\circ$)	Mean Trans. Err. ($^\circ$)	Median Trans. Err. ($^\circ$)
2	Classical	0.730	0.586	48.892	44.199
	SRPose	0.547	0.509	17.519	11.541
5	Classical	3.884	1.270	48.475	46.834
	SRPose	0.715	0.672	11.736	8.226
10	Classical	16.718	2.072	47.212	42.250
	SRPose	0.982	0.896	10.425	6.687
12	Classical	14.669	2.492	45.894	37.337
	SRPose	1.106	0.939	9.926	6.692

4.4 Error Distribution Analysis

We can clearly see from Figure 3 that at skip=2, classical method’s distribution is unimodal and comparable to SRPose’s; However, from skip=5 onward, the classical method shows two additional clusters, one near 90° and another near 180° , both of which are absent from SRPose’s distribution. This is tied directly to the mean/median divergence issue observed in Table 3, and hints at a cheirality-candidate confusion and failure.

4.5 Catastrophic Failure Characterization

To quantify the catastrophic failures across frame-skip values, we measured the number of image-pairs that gave near- 90° and near- 180° rotation errors in the classical method.

Table 4 shows no failures at skip = 2, consistent with the unimodal error distribution in Figure 3. Failure counts appear and increase with the skip value, with 2 Near- 180° failures at skip = 5 rising to 12 at skip = 10, then dropping slightly to 8 failed pairs at skip = 12, and a steady increase of Near- 90° failures from 1 failure at skip = 2, to 8 at skip = 5, and reaching 10 failures at the last skip. We can notice also that as the skip value increases, the average true rotation magnitude increases as well between the two failed image pairs.

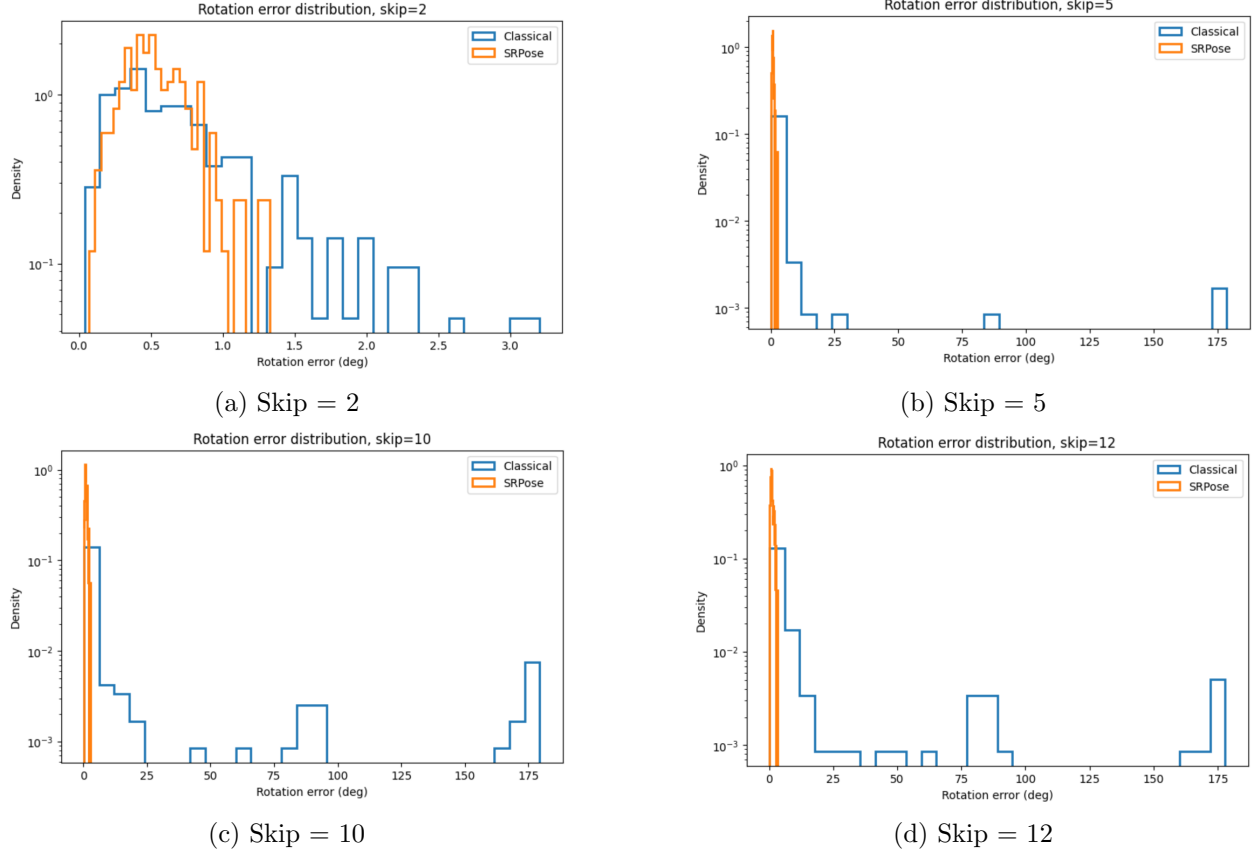


Figure 3: Distribution of rotation error for the classical pipeline (blue) versus SRPose (orange) across increasing frame-skip values.

Table 5 reports the fraction of pairs exceeding a fixed error threshold for each method, by skip value: 90° for the classical pipeline, corresponding to catastrophic cheirality-disambiguation failure, and 1° for SRPose, corresponding to departure from its typical sub-degree precision. Both fractions increase with skip value, though not strictly monotonically for the classical pipeline. SRPose’s fraction exceeding its 1° threshold increases monotonically and substantially, from 0.045 at skip = 2 to 0.465 at skip = 12, showing that while SRPose avoids catastrophic errors entirely at every tested skip value, its typical precision also degrades as frame-skip increases.

Table 4: Number of catastrophic (near- 180°) and secondary (near- 90°) failures, by frame-skip value, with mean ground-truth rotation magnitude of each failure cluster.

Skip	# Near- 180°	Mean r_{gt} ($^\circ$)	# Near- 90°	Mean r_{gt} ($^\circ$)
2	0	—	0	—
5	2	6.53	1	9.68
10	12	15.02	8	16.04
12	8	20.11	10	22.50

Table 5: Fraction of pairs exceeding a fixed error threshold, by frame-skip value, for the classical pipeline (threshold: 90°) and SRPose (threshold: 1°).

Skip	Classical: frac. rot. err. > 90	SRPose: frac. rot. err. > 1
2	0.000	0.045
5	0.010	0.166
10	0.071	0.379
12	0.045	0.465

4.6 Case Study: Skip=10

In this section, we will look closely at skip value = 10 since it had a substantial number of near-180° failures (12) and near-90° failures (8), giving enough cases for meaningful subgroup analysis.

4.6.1 Example of Failed Image Pairs

To inspect the failed cases, we focused on the 3 worst near-180° image pairs in skip value = 10, and noted their inlier count, the match coverage, the baseline and true rotation magnitude between the pair, the singular values of the Essential matrix, as well as the cheirality test total scores of each of the 4 (R,t) candidates and the reprojection errors.

All three pairs in Table 6 show a severe Essential Matrix degeneracy with singular value ratios around 1 : 649 for pair 805, 1 : 1754 for pair 845 and 1 : 3.59 for pair 190.

Pairs 805 and 845 both display a very small baseline, a moderate ground truth rotation magnitude, and a limited match coverage (31 and 41 inliers covering around 25% and 28% of the scene, respectively), resulting in poor reprojection fit across all 4 (R,t) candidates.

Pair 190 particularly displays an even worse coverage with 10 inliers only, covering just 7.8% of the scene. This can also be observed in Figure 4c, where we can see about 8-10 matched points clustered together in one small spot of the scene.

Table 6: Case studies of catastrophic (near-180°) rotation errors at skip = 10.

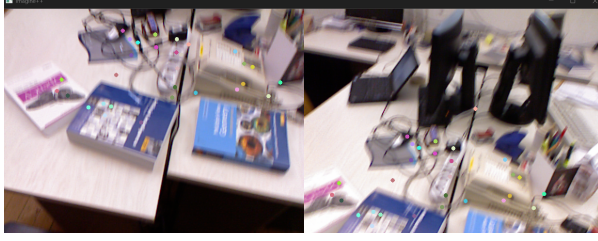
Pair	r_{gt} (°)	Baseline	Inliers	Coverage	E Singular Values	Cheirality Totals	Reproj. Err. (R_1 / R_2 , px)	Rot. Err. (°)
805	19.33	0.0142	31	0.250	0.999, 0.00154, ~ 0	22 / 0 / 21 / 0	181.9 / 139.0	179.508
845	16.84	0.0180	41	0.280	1.000, 0.00057, ~ 0	37 / 0 / 29 / 0	305.3 / 425.2	179.085
190	18.19	0.0139	10	0.078	0.963, 0.268, ~ 0	0 / 7 / 0 / 4	49.1 / 767.6	178.560

4.6.2 Inliers and Coverage

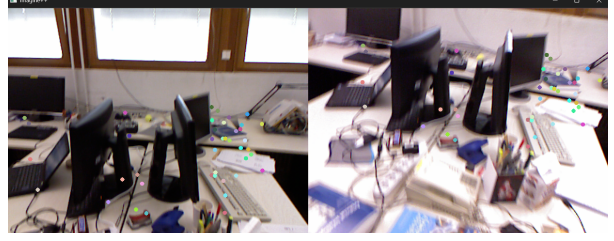
To test whether the number and the spread of matches in the failed image pairs directly affect the rotation error, we analyzed the inlier count and coverage percentage across all pairs for skip value = 10. We can see in Table 7 that failed pairs have substantially fewer inliers (47.50 vs 110.35) and lower coverage (27% vs 39%) than non-failed pairs, on average.

The same analysis was done on skip values 5 and 10, can be found in the appendix E.4, and show similar patterns.

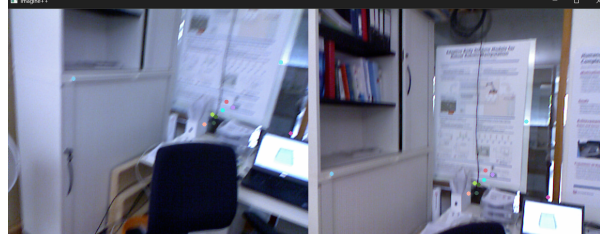
Finally, we plotted the inlier count and the match coverage’s evolution with the increase of the skip value, and we can see in Figure 5 that as the skip value increases, three things move together: rotation magnitude rises, the inlier count falls, and the coverage falls.



(a) Pair 805 ($r_{gt} = 19.33^\circ$, 31 inliers)



(b) Pair 845 ($r_{gt} = 16.84^\circ$, 41 inliers)



(c) Pair 190 ($r_{gt} = 18.19^\circ$, 10 inliers)

Figure 4: Matched keypoints (colored dots, matching color indicates correspondence) for three catastrophic-failure case studies at skip = 10.

Table 7: Mean characteristics of failed (rotation error $> 150^\circ$) versus non-failed pairs at skip = 10.

	All pairs	Failed	Non-failed
r_{gt} magnitude ($^\circ$)	11.41	15.02	11.18
Inliers	106.54	47.50	110.35
Match coverage	0.38	0.27	0.39

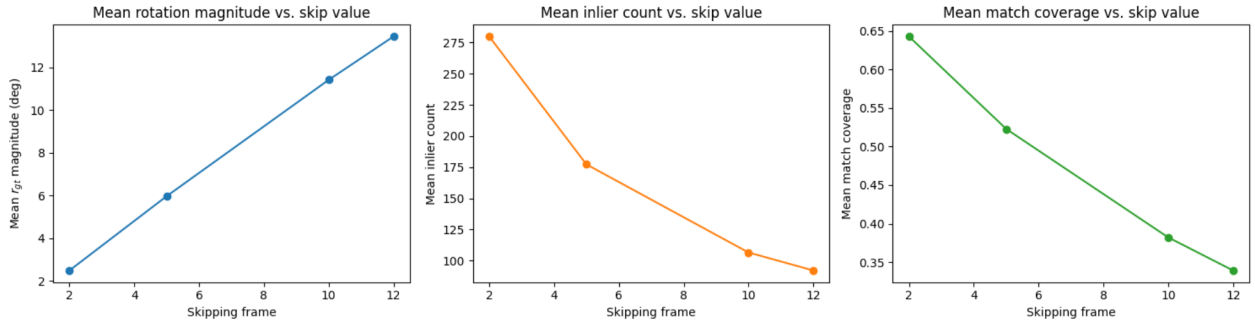


Figure 5: Mean ground-truth rotation magnitude (left), mean inlier count (center), and mean spatial match coverage (right) as a function of frame-skip value.

4.7 Case Study: Skip=2

To compare the performance of the classical method and SRPose on non-catastrophic cases, we focus on skip value = 2 for which the classical method had 0 failures.

Figure 6 compares classical and SRPose rotation error on a per-pair basis to see whether classical’s worse pairs are the same pairs where SRPose also does worse. The majority of points lie above the $y=x$ line, indicating that classical error exceeds SRPose error on most individual pairs. This excess is distributed broadly across the range of SRPose error values (approximately 0.1° – 1.3°) rather than concentrated at any particular difficulty level, and a small minority of pairs (bottom-right region) show the reverse pattern, with the classical method outperforming SRPose.

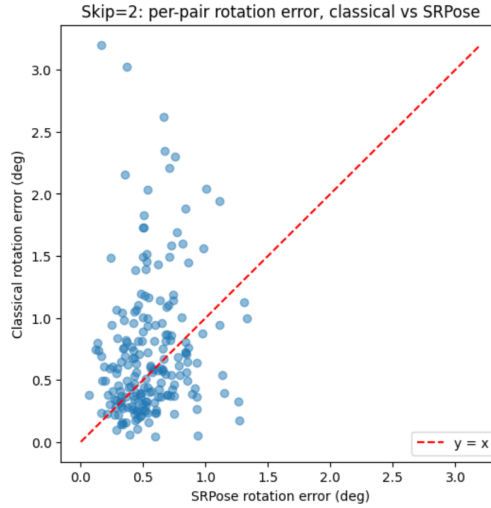


Figure 6: Per-pair rotation error comparison between the classical pipeline and SRPose at skip = 2.

Table 8: Best-case rotation error (top 3%, mean and max) and overall standard deviation of rotation error for the classical pipeline versus SRPose, at skip = 2.

Method	Top 3% Mean ($^{\circ}$)	Top 3% Max ($^{\circ}$)	Std. Dev., All Pairs ($^{\circ}$)
Classical	0.0812	0.1398	0.556
SRPose	0.1316	0.1625	0.242

Finally, Table 8 reports the mean and maximum rotation error within the top 3% best-performing pairs (that reported the lowest rotation errors), together with the overall standard deviation, for both methods at skip = 2. The classical pipeline achieves a lower mean (0.0812° vs. 0.1316°) and maximum (0.1398° vs. 0.1625°) rotation error than SRPose within this best-case subset, while its overall standard deviation (0.556°) is more than double that of SRPose (0.242°) across the full set of 200 pairs.

4.8 All Frames

Similarly to skip value = 2, we show the top 3% best-performing pairs (mean and maximum rotation error) and the overall standard deviation of the rotation error for the classical method and

for SRPose across the remaining skip values (5, 10, and 12) in Table 9, where we can also see that the classical pipeline achieved a lower top 3% mean rotation error than SRPose in all cases (and a lower top 1% maximum at skip = 5 and 10), while having its standard deviation be remarkably higher than that of SRPose, especially as the skip value increases.

Table 9: Best-case rotation error (top 3%, mean and max) and overall standard deviation of rotation error for the classical pipeline versus SRPose, at skip = 5, 10, and 12.

Skip	Method	Top 3% Mean (°)	Top 3% Max (°)	Std. Dev., All Pairs (°)
5	Classical	0.1163	0.2073	18.595
	SRPose	0.1834	0.2202	0.353
10	Classical	0.2158	0.2627	43.771
	SRPose	0.2296	0.3151	0.502
12	Classical	0.1651	0.3119	37.346
	SRPose	0.2549	0.3009	0.608

5 Discussion

5.1 Why AC-RANSAC Outperforms Classical RANSAC

The improvement observed when replacing classical RANSAC with AC-RANSAC (ORSA) can be attributed to some extent to the fundamental methodological difference between the two approaches: unlike classical RANSAC, AC-RANSAC does not require a fixed inlier/outlier threshold (`distMax`) to be specified in advance. Instead, it automatically determines an appropriate threshold for each image pair individually, using an a-contrario (NFA-based) criterion that selects the model and precision jointly. In RANSAC, the appropriate threshold for a given pair depends on factors that vary substantially from one image pair to the next (like scene structure, correspondence noise level, baseline). This cannot be covered by a single global constant, and this is where AC-RANSAC’s data-adaptive threshold selection overcomes this limitation.

5.2 Divergence Between Mean and Median Error in the Classical Pipeline

Before comparing the classical pipeline against SRPose, it is important to note that translation error was substantially larger than rotation error for both methods throughout this evaluation, which could be attributed to the difficulty and instability in recovering the translation direction in frames where the translation is extremely small. Therefore, we focus specifically on rotation error for the remainder of this discussion, which is more directly diagnostic of the pose decomposition and disambiguation failures under investigation.

When it comes to the performance of the classical pipeline, we observed a substantial divergence between the mean and median values of both rotation and translation error at every skip value greater than 2, a pattern absent from SRPose’s results. This divergence is indicative of a small number of extreme outliers pulling the mean upward without materially affecting the median. This hypothesis was directly confirmed by the rotation error histograms (Section 4.4), which revealed that, beyond skip value = 2, the classical pipeline’s error distribution is not unimodal, but instead exhibits secondary clusters near 90° and 180°. These secondary clusters correspond to cases in which the essential matrix decomposition selected an incorrect rotation/translation candidate during the cheirality-based test, rather than cases of moderately degraded precision.

Upon inspecting these failed pairs individually (Section 4.6.1), we consistently observed severe degeneracy in the estimated essential matrix, evidenced by singular value ratios far from the theoretically expected values of a well-conditioned essential matrix.

5.3 Inliers, Match Coverage, and Rotation Magnitude in Catastrophic Failures

Examining the near-180° failure cases of the classical method in more detail, we found that a low inlier count and low spatial coverage of the matched correspondences over the scene were the most dominant and common factors shared among these cases. A direct example of this would be pair number 190 4, where we can see that the inliers cluster on a flat surface of the scene, leading to a poor essential matrix conditioning caused by the lack of depth information.

Compared to non-failed pairs, failed pairs consistently exhibited fewer inliers and lower coverage on average (Table 7), even after accounting for the fact that coverage and inlier count both declined with increasing skip value across the dataset as a whole (Figure 5). That is, the gap between failed and non-failed pairs was not simply a byproduct of overall data degradation at higher skip.

We also observed that the mean ground-truth rotation magnitude of the failing population tended to be higher than that of the non-failed population. By visual inspection of the failed image pairs in figure 4, we can see that these pairs exhibit substantial viewpoint change between the two images. This suggests that it is not only the magnitude of the relative rotation and translation that governs estimation difficulty, but also their direction relative to the scene. This viewpoint change is plausibly also responsible for the broader decline in inlier count and coverage observed as the skip value increases: a larger gap between frames allows more time for the camera to move through a trajectory that reduces the shared field of view between the two images, regardless of the specific rotation magnitude involved.

5.4 Why SRPose Avoids Catastrophic Failures

The absence of catastrophic rotation errors in SRPose’s results, even at higher skip values, is plausibly attributable to an architectural difference between the two methods. Rather than relying on a discrete set of hard correspondences established through explicit matching, SRPose instead uses keypoint descriptor similarity as a bias term within its cross-attention layers, as explained in Section 2.3.1. This allows the model to make use of information from all keypoints detected in both images jointly, rather than committing to a fixed subset of point-to-point matches before pose estimation begins. As a result, SRPose benefits from a richer, more geometrically informative representation of the scene. It thus removes the risk inherent to the classical pipeline’s explicit matching stage, where a small number of incorrect or poorly distributed correspondences can propagate directly into a degenerate essential matrix estimate and ultimately a catastrophically wrong pose.

5.5 Precision Versus Robustness

Finally, to carefully examine the precision and robustness of our two methods, we examined skip value = 2 in particular, where the classical pipeline exhibited no catastrophic failures. The lack of failed cases could partly be attributed to the small motion between the chosen frames, giving us better inlier coverage and also a rotation matrix relatively closer to the identity matrix, perhaps making the estimation process a less challenging one. However, upon closer inspection of the rotation errors, the per-pair comparison in Figure 6 shows that the majority of pairs lie above the $y = x$ line, indicating higher classical error than SRPose on most individual pairs. This pattern suggests that the classical method’s excess error at skip = 2 is not primarily explained by scene or

pair-specific difficulty, but rather reflects an additional source of noise specific to the pipeline (like SIFT keypoint localization imprecision or residual limitations of the FNS noise model), layered on top of whatever intrinsic difficulty a given pair presents to either method.

At the same time, the small cluster of points falling below the $y = x$ line in Figure 6 (pairs where the classical approach achieved lower error than SRPose), indicates that the classical method is not uniformly worse. This behavior is also reflected in the remaining skip values (Figure 13), where we also had pairs perform better through the classical pipeline.

This was also seen more directly in the recorded top 3% best-performing pairs at each skip value (Tables 8, 9), where the classical pipeline’s top 3% mean rotation error was in fact lower than SRPose’s top 3% mean at every skip value, showing that classical estimation is capable of exceeding SRPose’s best-case precision when conditions are favorable; a similar advantage was observed for the top 3% maximum rotation error at three of the four skip values, with SRPose showing a marginally lower maximum only at skip = 12. This is further reinforced by our synthetic validation results (Section 4.1), in which the classical pipeline recovered the ground-truth essential matrix within machine precision on noiseless data. This indicates that the classical pipeline has no inherent floor on its achievable precision and that its limit in practice arises not from the method itself, but from the dataset and from the fundamental nature of a two-dimensional, hard-correspondence-based approach, which must compete against a deep learning method, with no hard-correspondences, exposed during training to a wide variety of scenes and viewing conditions. Despite these limitations, we can say that its precision competes with that of SRPose’s, exceeding it in some cases.

On the other hand, when it comes to robustness, the classical pipeline’s standard deviation of rotation errors always exceeded that of SRPose, and for skip value = 2 in particular, where no catastrophic pairs exist, it was more than double SRPose’s. This indicates that the favorable best-case performance of the classical approach is realized inconsistently, and that SRPose was consistently more robust throughout our experiments, maintaining stable, predictable accuracy across a substantially wider range of scene conditions and frame-skip values.

Together, these results characterize a precision-versus-robustness trade-off: classical geometric estimation can achieve superior peak accuracy when correspondence quality and scene geometry are favorable, but SRPose provides more reliable, consistent performance across the full range of tested conditions.

6 Limitations and Future Work

6.1 Limitations of the Classical Pipeline

One limitation of our classical method lies in the noise model assumed by Kanatani’s fundamental matrix estimation: it assumes isotropic, identical-variance noise across both images, treating the coordinate perturbations of x , y , x' , and y' as independent random variables with equal variance σ^2 . If SIFT keypoint localization noise differs between the two images of a pair (for instance, due to differing local texture, blur, or illumination) this assumption may not hold. Additionally, we used Kanatani’s FNS as the primary maximum-likelihood-style refinement method, with a Gauss-Newton-based estimator implemented as a secondary point of comparison. However, a full comparative evaluation of the two was not carried out. Future work should evaluate the classical pipeline using alternative estimators including Gauss-Newton, HEIV and iterative renormalization to determine whether they could offer improved accuracy or robustness relative to FNS on this dataset.

Furthermore, we also noticed the pipeline’s sensitivity to Lowe’s Ratio Threshold. We found that relaxing the threshold could substantially fix some catastrophic rotation errors by increasing the number of correspondences, but could worsen average performance across the dataset. This indicates that its effect is highly case-dependent, reflecting an underlying bias-variance tradeoff. Future work could explore an adaptive ratio threshold per image pair rather than the single fixed threshold used throughout this work.

Finally, the classical pipeline, as implemented, is considerably slower to run than would be desirable for practical use. Future work should time each major stage of the pipeline to identify which stage dominates total running time and to guide targeted optimization efforts.

6.2 Other Limitations

The nearest-timestamp association performed between RGB frames and ground-truth pose entries may also be a limiting factor on the precision of either method. Any residual synchronization error introduced by this association places a floor on the smallest measurable error, independent of either pipeline’s true underlying accuracy. We also note that this work did not focus in depth on translation error, which was substantially larger than rotation error for both methods, and markedly worse for the classical pipeline. A thorough investigation of translation error is left to future work.

Moreover, as discussed in Section 2.3.1, SRPose applies a linear intrinsic normalization to key-point coordinates. However, there is no indication of explicit correction for lens distortion, unlike the correction done in the classical pipeline. We noted that at least one of its training datasets (MegaDepth) is documented to be fully undistorted prior to training, but other training datasets, such as ScanNet, are released with raw, distorted RGB imagery. Consequently, it remains unclear and left for future work whether this heterogeneity affects the model’s implicit handling of distortion at inference time on the TUM RGB-D imagery used in this work.

Finally, we had originally intended to evaluate the precision of both methods under controlled, pixel-injected noise, in order to test the precision-versus-robustness tradeoff more directly; this experiment was not completed within the scope of this work. Also, we wished to evaluate both methods’ precision on real image content under fully controlled ground truth, without the confounding real-world factors (such as motion blur). This could be achieved by taking a single high-quality image, applying a known homography to it, and feeding the original and transformed images to both the classical pipeline and SRPose. This approach would isolate each method’s precision under exact ground truth on real image content, complementing both the purely synthetic validation and the real, but less controlled, TUM RGB-D evaluation presented in this work.

7 Conclusion

This work built a classical relative pose estimation pipeline from first principles, validated its correctness via synthetic ground-truth testing, and compared it against SRPose, a state-of-the-art learned method, on real TUM RGB-D data. The comparison revealed a nuanced result: SRPose consistently achieved lower mean and median error and remained stable as frame-skip increased, while the classical pipeline performed competitively at low skip but showed a growing rate of catastrophic rotation errors at higher skip, traced primarily to poor correspondence coverage, a vulnerability SRPose largely avoids by drawing on a richer, attention-weighted scene representation rather than a discrete set of hard correspondences. Despite this, the classical pipeline achieved a lower average rotation error than SRPose among the top 3% best-performing pairs at every skip value, characterizing a precision-versus-robustness tradeoff rather than a simple superiority of one

method over the other. This work identifies specifically when, how, and why the classical pipeline fails and SRPose outperforms it, and when it does not, with the identified limitations offering concrete directions for sharpening this picture further.

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The linear algebra utilities (matrices and vectors), as well as the RANSAC and AC-RANSAC implementations used in this work, are based on the `libOrsa` library (which incorporates `libNumerics`), developed by Prof. Pascal Monasse. This code was obtained directly from the IPOL article [12].

Ground-truth trajectories, camera calibration parameters, and the `associate.py` association script for the TUM RGB-D dataset were obtained from the official TUM RGB-D dataset website.¹ SRPose was evaluated using the official implementation released by its authors.²

The pipeline diagram presented in Figure 2 was generated with the assistance of ChatGPT, based on a description and structure provided by the author. We would like to also acknowledge the use of Claude (Anthropic) for assistance with C++ syntax questions and debugging during the implementation of this work.

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¹https://cvg.cit.tum.de/data/datasets/rgbd-dataset/file_formats

²<https://github.com/frickyinn/SRPose>

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A Full Derivation of Kanatani’s Maximum Likelihood Estimator

This appendix presents the complete derivation of Kanatani’s maximum likelihood fundamental matrix estimator, summarized in Section 2.2.3.

A.1 Noise Propagation

In the one-dimensional case, $\text{Var}(x) = \mathbb{E}[(x - \bar{x})^2]$, and analogously

$$\text{Var}(\xi_\alpha) = \mathbb{E}[(\xi_\alpha - \bar{\xi}_\alpha)^2], \quad \xi_\alpha - \bar{\xi}_\alpha = \Delta\xi_\alpha. \quad (8)$$

For the higher-dimensional case, this gives

$$\text{Var}(\xi_\alpha) = \mathbb{E} \left[\Delta \xi_\alpha \Delta \xi_\alpha^\top \right] = V[\xi_\alpha]. \quad (9)$$

Using Jacobian error propagation,

$$\text{Cov}(\xi_{\alpha_i}, \xi_{\alpha_j}) \approx \sigma^2 \left(\frac{\partial \xi_i}{\partial x} \frac{\partial \xi_j}{\partial x} + \frac{\partial \xi_i}{\partial y} \frac{\partial \xi_j}{\partial y} + \frac{\partial \xi_i}{\partial x'} \frac{\partial \xi_j}{\partial x'} + \frac{\partial \xi_i}{\partial y'} \frac{\partial \xi_j}{\partial y'} \right), \quad (10)$$

so that $V[\xi_\alpha]$ can be decomposed as $V[\xi_\alpha] = \sigma^2 V_0[\xi_\alpha]$, with

$$V_0[\xi_\alpha] = \begin{pmatrix} \bar{x}_\alpha^2 + \bar{x}'_\alpha{}^2 & \bar{x}_\alpha \bar{y}_\alpha & f_0 \bar{x}_\alpha & \bar{x}_\alpha \bar{y}'_\alpha & 0 & 0 & f_0 \bar{x}'_\alpha & 0 & 0 \\ \bar{x}_\alpha \bar{y}_\alpha & \bar{x}_\alpha^2 + \bar{y}'_\alpha{}^2 & f_0 \bar{y}_\alpha & 0 & \bar{x}_\alpha \bar{y}'_\alpha & 0 & 0 & f_0 \bar{x}_\alpha & 0 \\ f_0 \bar{x}_\alpha & f_0 \bar{y}_\alpha & f_0^2 & 0 & 0 & 0 & 0 & 0 & 0 \\ \bar{x}_\alpha \bar{y}'_\alpha & 0 & 0 & \bar{y}_\alpha^2 + \bar{x}'_\alpha{}^2 & \bar{x}'_\alpha \bar{y}'_\alpha & f_0 \bar{x}'_\alpha & f_0 \bar{y}_\alpha & 0 & 0 \\ 0 & \bar{x}_\alpha \bar{y}'_\alpha & 0 & \bar{x}'_\alpha \bar{y}'_\alpha & \bar{y}_\alpha^2 + \bar{y}'_\alpha{}^2 & f_0 \bar{y}'_\alpha & 0 & f_0 \bar{y}_\alpha & 0 \\ 0 & 0 & 0 & f_0 \bar{x}'_\alpha & f_0 \bar{y}'_\alpha & f_0^2 & 0 & 0 & 0 \\ f_0 \bar{x}'_\alpha & 0 & 0 & f_0 \bar{y}_\alpha & 0 & 0 & f_0^2 & 0 & 0 \\ 0 & f_0 \bar{x}_\alpha & 0 & 0 & f_0 \bar{y}_\alpha & 0 & 0 & f_0^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (11)$$

V_0 depends only on the true coordinates and the conditioning constant f_0 ; the covariance weighting it encodes determines how much influence each measurement's error has on the final estimate. The zero entries correspond to terms of ξ_α that are independent of the coordinates x, x', y, y' and are therefore noise-free, with no associated fluctuation or variance. In $V[\xi_\alpha]$, σ^2 represents the magnitude of the noise, while V_0 represents its structure. For example, the uncertainty of the first term of ξ_α ($x x'$) is $\bar{x}_\alpha^2 + \bar{x}'_\alpha{}^2$, meaning that points farther from the image center exhibit greater coordinate uncertainty even under uniform pixel-level noise σ . $V_0[\xi_\alpha]$ thus acts as a per-correspondence weighting profile for statistical normalization: a small covariance indicates a highly informative measurement, while a large covariance indicates a weak or unreliable one, and each correspondence's contribution to the estimation of u is scaled accordingly.

To quantify the uncertainty of an estimate $\hat{u}$, only the component of $\hat{u}$ perpendicular to the true u is considered, since the parallel component is fixed by the constraint $\|u\| = 1$. This is achieved by projecting $\hat{u}$ with the matrix P_u , which projects any vector onto the direction orthogonal to u ; $\|P_u \hat{u}\|$ then measures the deviation of the estimate from the true u in the meaningful, orthogonal direction only.

A.2 Lagrangian Derivation

To eliminate the constraint $(u, \bar{\xi}_\alpha) = 0$ from Eq. 4, a Lagrange multiplier is introduced, converting the problem into an unconstrained minimization. The Lagrangian for a single correspondence is

$$L(\bar{\xi}_\alpha, \lambda_\alpha) = \frac{1}{2} (\xi_\alpha - \bar{\xi}_\alpha, V_0^-(\xi_\alpha - \bar{\xi}_\alpha)) - \lambda_\alpha (u, \bar{\xi}_\alpha), \quad (12)$$

with λ_α a scalar multiplier. Setting $\partial L / \partial \bar{\xi}_\alpha = 0$ gives

$$-V_0^-(\xi_\alpha - \bar{\xi}_\alpha) - \lambda_\alpha u = 0 \implies \xi_\alpha - \bar{\xi}_\alpha = -\lambda_\alpha V_0 u, \quad (13)$$

from which

$$\lambda_\alpha = -\frac{(u, \xi_\alpha)}{(u, V_0 u)}, \quad \xi_\alpha - \bar{\xi}_\alpha = \frac{(u, \xi_\alpha)}{(u, V_0 u)} V_0 u. \quad (14)$$

Substituting Eq. 14 back into J_{ML} (Eq. 4) yields the reduced cost function of Eq. 5.

A.3 Interpretation of M and L

In J_{ML} (Eq. 5), the numerator of each term measures the epipolar constraint error, while the denominator incorporates the covariance, that is, the uncertainty associated with that constraint. Rather than measuring error in a fixed Euclidean metric, this formulation measures error in a metric that depends on the correspondence noise model V_0 , pulled back through the epipolar constraint. In the resulting stationarity condition $(M - L)u = 0$, Mu aggregates constraint-aligned directions across all correspondences, while Lu aggregates uncertainty-weighted curvature corrections; M can be interpreted as the weighted data moment (the weighted distribution of the correspondence data), while L represents a bias correction induced by the noise structure.

A.4 FNS Algorithm and Initialization

Algorithm 1 summarizes the FNS iteration used in this work. It was later showed that convergence performance improves if the eigenvalue closest to zero in Step 3 is replaced with the algebraically smallest eigenvalue, a variant referred to as *modified FNS* [8].

Algorithm 1 FNS Algorithm for Fundamental Matrix u Estimation

- 1: Initialize $u_{\text{old}} \leftarrow u_0$
 - 2: **repeat**
 - 3: Compute $M(u_{\text{old}})$ and $L(u_{\text{old}})$
 - 4: Solve the problem $(M - L)u_{\text{new}} = 0$ which corresponds to solving the eigenvalue problem: $(M - L)u_{\text{new}} = \lambda u_{\text{new}}$: we find eigenvector u_{new} that has eigenvalue λ closest to 0
 - 5: Update $u_{\text{old}} \leftarrow u_{\text{new}}$
 - 6: **until** $u_{\text{new}} \approx u_{\text{old}}$
 - 7: **return** u_{new}
-

Alternative estimators. Several other estimators address the same estimation problem. The **HEIV** (Heteroscedastic Errors-in-Variables) method [8] and iterative **renormalization** [8] both aim to solve the same maximum-likelihood-style optimality condition as FNS, differing primarily in their iterative update schemes and convergence properties, while **Gauss-Newton**-based approaches directly minimize a reprojection- or Sampson-error-based nonlinear cost function via local linearization and iterative refinement.

Since FNS is a local iterative refinement method, its convergence depends on the quality of its initial estimate u_{init} . In this work, the fundamental matrix returned by AC-RANSAC is used to construct u_{init} , after normalizing it to unit Frobenius norm and converting it from the standard pixel-space convention into Kanatani’s f_0 -conditioned convention. This provides FNS with a geometrically sound starting point, already free of gross outliers, which was found in this work to be necessary for reliable convergence to the correct stationary point of J_{ML} .

B SRPose Attention and Loss Function Details

B.1 Self-attention module:

I_1 's embeddings X_1 are used to compute a self-attention score of its descriptors with themselves: $A_s = \frac{qk^\top}{\sqrt{d}}$, where q and k are the queries and keys obtained from the embeddings of I_1 . Applying softmax to A_s and multiplying the result by v (also obtained from the embeddings of I_1) yields the self-attention module embeddings X_1^s . The same procedure is repeated for I_2 to obtain X_2^s . Cross-attention: Keys and values are computed again, this time from the outputs of the self-

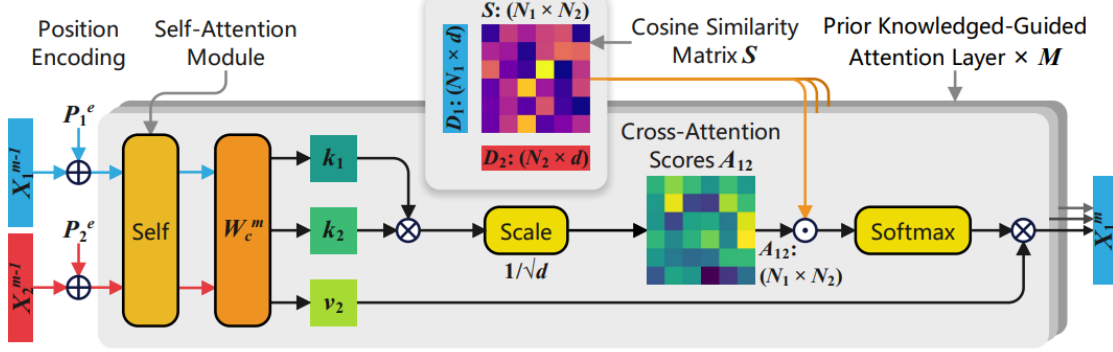


Figure 7: SRPose Attention Layer[17]

attention module, X_1^s and X_2^s , giving k_1 , k_2 , v_1 , and v_2 . Cross-attention is then computed as $A_{12} = A_{21} = \frac{k_1 k_2^\top}{\sqrt{d}}$.

B.2 Loss function:

SRPose is trained under full supervision, with the total loss given by $\mathcal{L}(I_1, I_2) = \mathcal{L}_R + \lambda_t \mathcal{L}_t + \lambda_{tn} \mathcal{L}_{tn} + \lambda_{ta} \mathcal{L}_{ta}$, where $\mathcal{L}_R$ is the rotation loss, computed as the Huber loss of the error in the rotation angle. Since translation direction alone is unrecoverable in scale from a monocular pair, the translation error is instead supervised via three complementary angular losses: $\mathcal{L}_t$, the angular error between the predicted and ground-truth translation directions; $\mathcal{L}_{tn}$, an angular loss computed on the translation vector normalized within a local or batch-relative scale, encouraging the correct direction independent of absolute scale; and $\mathcal{L}_{ta}$, an auxiliary angular loss over an alternative or averaged translation representation, further regularizing the direction estimate [17].

C Duplicate Match Removal

In the Implementation step of the classical pipeline and following the undistortion process of the correspondences extracted by SIFT, the resulting set of correspondences is then filtered to remove exact or near-exact duplicate matches, which were matches that shared the same coordinates up to a small numerical tolerance. This was done in order to prevent a single physical match to have a much bigger influence than it should on the RANSAC inlier filtering process.

D Numerical Considerations in The Implementation

We would like to note that singular value decomposition throughout the classical pipeline was performed using the Eigen linear algebra library [5] instead of the LibNumerics’ SVD [12] since it showed a much better numerical stability. Moreover, all random sampling performed by the pipeline’s own RANSAC implementation is seeded with a fixed value to ensure reproducibility.

E Additional Results

E.1 Skip = 10 Inliers and Coverage

Figure 8 shows that the image pairs having near-180° rotation errors all cluster below a threshold of 200 inliers and a 40% match coverage.

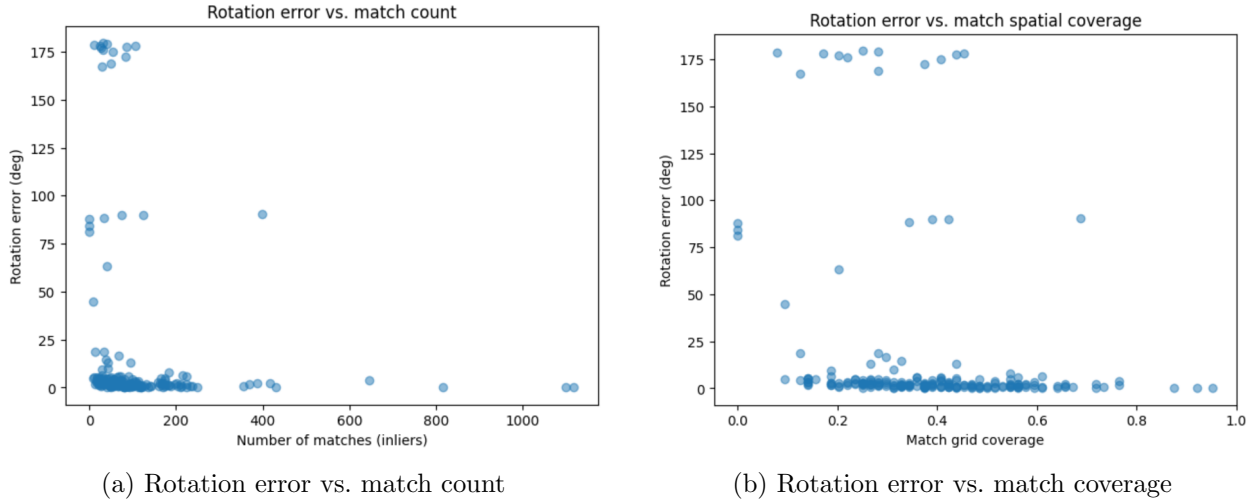


Figure 8: Rotation error as a function of match count (left) and spatial match coverage (right) at skip = 10.

E.2 Coverage Split of Failed Pairs at Skip = 10

Table 10: Failed pairs at skip = 10 split by coverage (low coverage = coverage < 20%), revealing two distinct failure profiles.

	r_{gt} magnitude (°)	Inliers	Coverage
Low-coverage failures	17.31	21.0	0.12
High-coverage failures	14.26	56.33	0.32

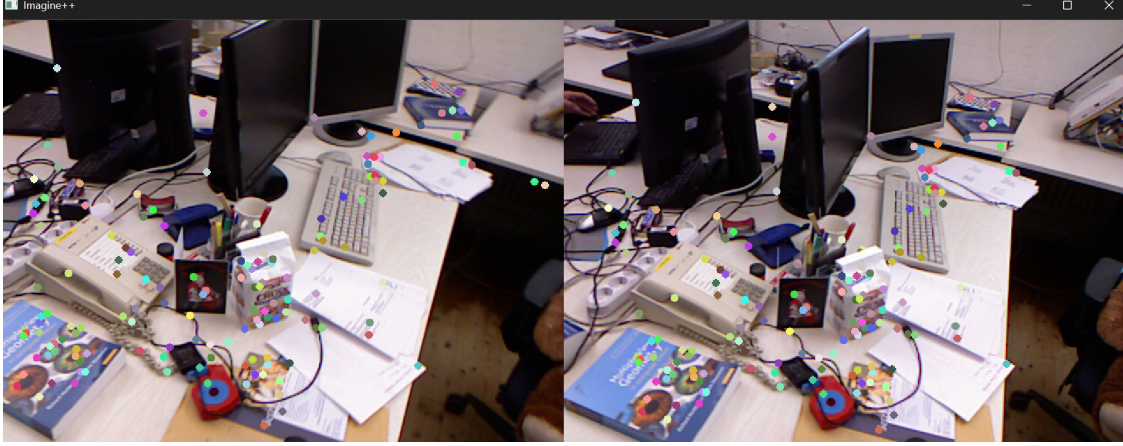
We can see from Table 10 that when we split the near-180° failures into high coverage ($\geq 20\%$) and low coverage ($< 20\%$) failures, both subgroups still displayed a lower inlier counts and a lower coverage relative to the non-failed average in Table 7 (110.35, 0.39).

E.3 Near-90 Failures: Baseline-Driven Degeneracy Independent of Correspondence Quality

Unlike near-180° failures (associated with poor inlier coverage), near-90° failures in this dataset are associated with extremely small baselines regardless of inlier count or coverage, evidenced most strikingly by pair 950 (398 inliers, 69% coverage, yet E singular value ratio $\approx 1850:1$).

Table 11: Case studies of secondary (near-90°) rotation errors.

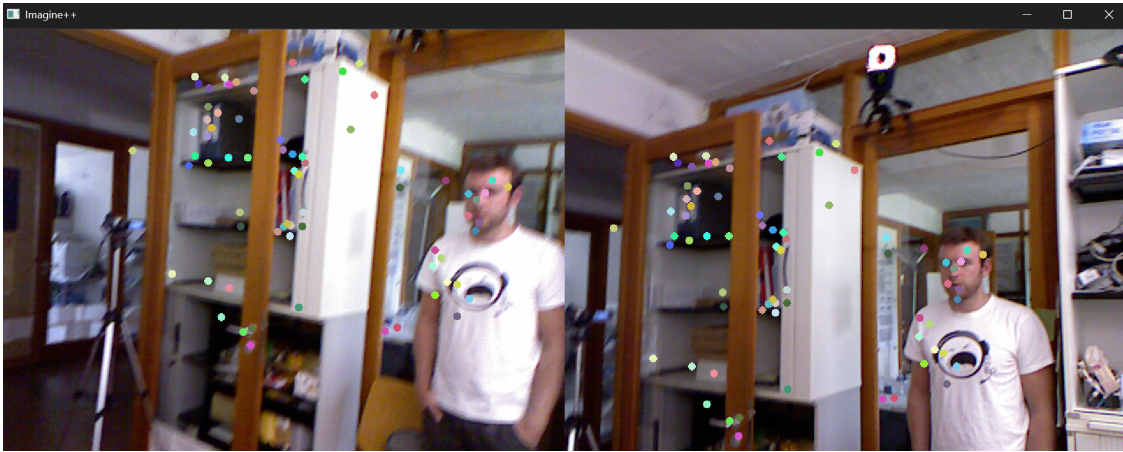
Pair (Image 1 / Image 2)	r_{gt} (°)	Baseline	Inliers	Coverage	E Singular Values	Cheirality Totals	Reproj. Err. (R_1 / R_2 , px)	Rot. Err. (°)
950	5.74	0.0232	398	0.688	1.000, 0.00054, ~ 0	27 / 54 / 5 / 2	410.0 / 315.5	90.511
905	10.76	0.0036	124	0.421	1.000, 0.00037, ~ 0	0 / 1 / 0 / 0	215.6 / 201.1	90.180
395	12.16	0.0104	74	0.390	1.000, 0.0000321, ~ 0	4 / 0 / 2 / 0	318.3 / 262.6	89.844



(a) Pair 950 ($r_{gt} = 5.74^\circ$, 398 inliers, 68.8% coverage)



(b) Pair 905 ($r_{gt} = 10.76^\circ$, 124 inliers, 42.1% coverage)



(c) Pair 395 ($r_{gt} = 12.16^\circ$, 74 inliers, 39.0% coverage)

Figure 9: Matched keypoints for three near-90° rotation error case studies.

A systematic comparison of baseline magnitude between near-90° failures ($n = 7$, mean = 0.104m) and non-failed pairs ($n = 177$, mean = 0.113m) at skip = 10 revealed no meaningful separation (Figure 10): the two distributions overlap substantially.

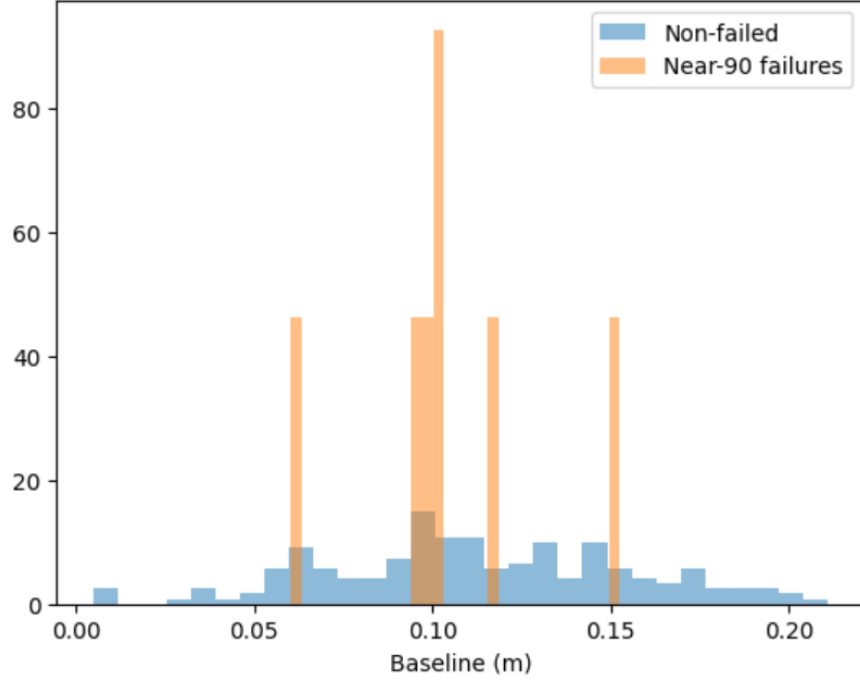


Figure 10: Distribution of baseline magnitude for near-90° failures versus non-failed pairs at skip = 10.

This indicates that baseline magnitude alone is not a reliable predictor of near-90° cheirality failures, and the mechanism underlying this failure mode remains only partially characterized. We note this as an open question for future investigation.

E.4 Skip Frames 5 and 12 Analysis

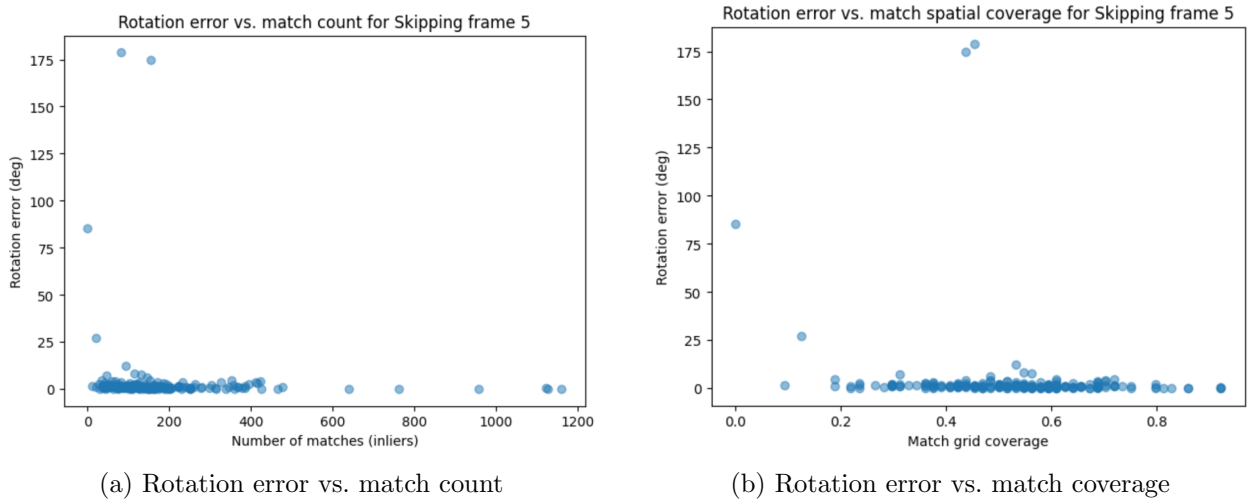


Figure 11: Rotation error as a function of match count (left) and spatial match coverage (right) at skip = 5.

Table 12: Mean characteristics of failed (rotation error $> 150^\circ$) versus non-failed pairs at skip = 5.

	All pairs	Failed	Non-failed
r_{gt} magnitude ($^\circ$)	5.96	6.53	5.96
Inliers	177.24	118.0	177.85
Match coverage	0.52	0.44	0.52

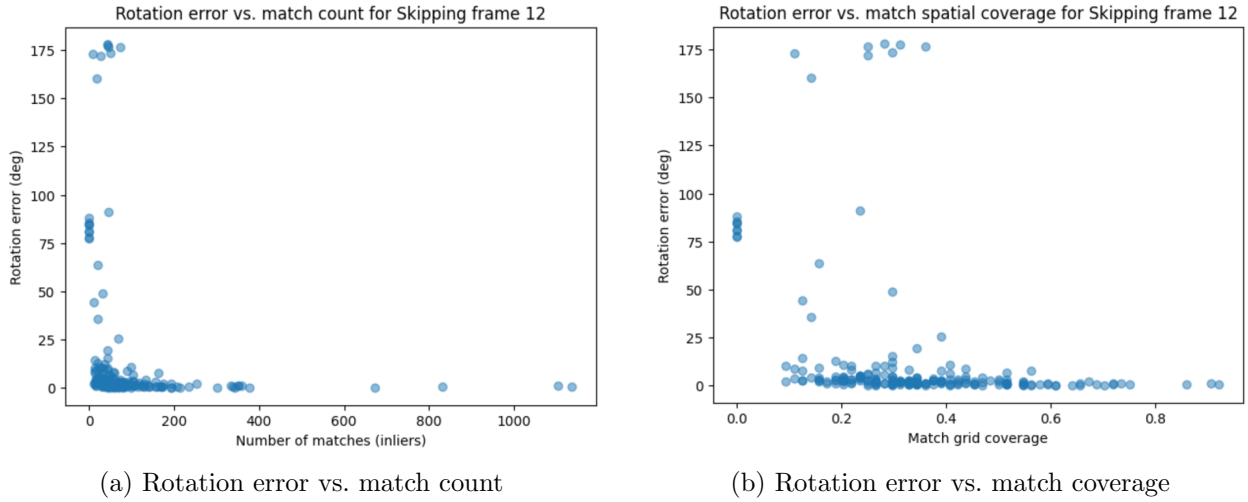


Figure 12: Rotation error as a function of match count (left) and spatial match coverage (right) at skip = 12.

Table 13: Mean characteristics of failed (rotation error $> 150^\circ$) versus non-failed pairs at skip = 12.

	All pairs	Failed	Non-failed
r_{gt} magnitude ($^\circ$)	13.44	20.10	13.16
Inliers	92.11	38.75	94.35
Match coverage	0.339	0.25	0.342

E.5 Per-Pair Comparison Across All Skip Values

Similarly to Figure 6, Figure 13 shows the comparison between classical and SRPose rotation error on a per-pair basis for all skip values, where we can also see a minority of pairs in the bottom right region showing the classical method outperforming SRPose for samples that had a rotation error below 5° .

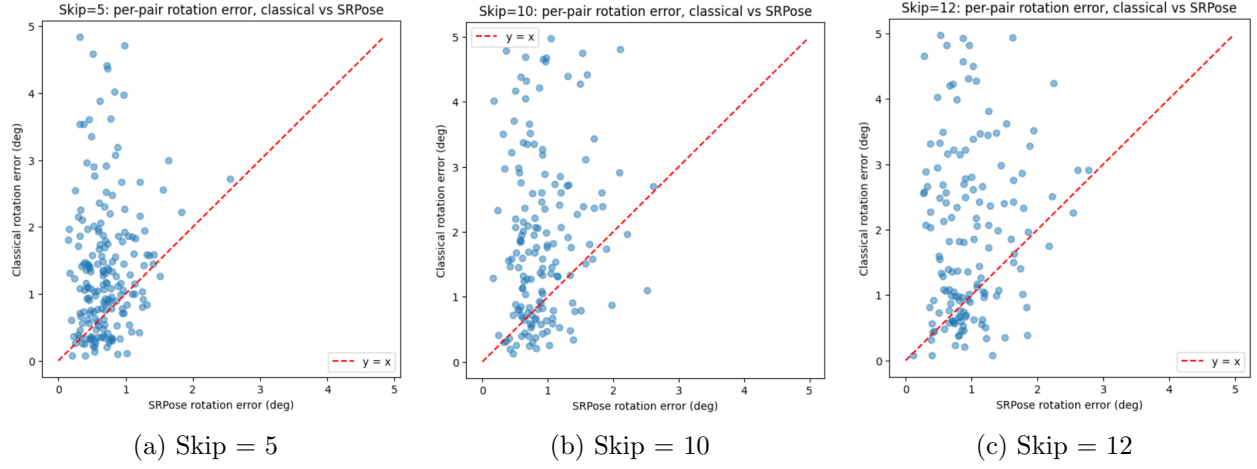


Figure 13: Per-pair rotation error comparison between the classical pipeline and SRPose at skip = 5, 10, and 12 for pairs having rotation error below 5° . The dashed line in each panel indicates equal performance ($y = x$).